$\pi_4(S^3) = \mathbb{Z}/2$ BY A POSTNIKOV UNIVERSAL PROPERTY: THE MAIN IDEAS OF A FORMAL PROOF

Attribution. David Wärn developed the universal-property strategy used here (one identifies a candidate Postnikov stage by a *marked universal property* and compares it with the truncated sphere on homotopy classes), together with the delooping constructions (Eilenberg–MacLane spaces as iterated torsors, and the criterion for a map of loop spaces to be the looping of a map [1]) and the cup-product construction. Our contribution was to develop the argument in detail and formalise it in Unit; the formal proof is the development `sag/pi3-s2-postnikov`, and the detailed notes are listed in §7. Owen Milner formulated the Sính invariant in homotopy type theory.

The result. In univalent type theory with replacement, and assuming only the universal property of S^3 (Axiom 1),

$$\pi_4(S^3) \cong \mathbb{Z}/2 \quad \text{as groups}$$

(`Pi4Sphere.Pi4S3.pi4-S3`, Theorem 8). No additional higher-inductive construction is assumed beyond the stated universal property of the sphere; no pushout, no smash product and no spectral sequence is used; the two homotopy-theoretic inputs are the cohomology computations (H3), (H4) of $K(\mathbb{Z}, 2)$, proved here from the circle and the cup square.

1. Setting

The foundations are those of `sag/em`: univalence and Rijke's replacement (images and their surjectivity) as parameters; function extensionality, propositional truncation, set quotients and the finite truncations $\|X\|_m$ are derived. Eilenberg–MacLane spaces $K(A, n)$ of an abelian group A are built as in [1]: $K(A, 1)$ is the type of A-torsors and $K(A, n{+}1)$ is the type of $K(A, n)$-torsors, i.e. of merely inhabited types with a homogeneous structure. We write $e_n : \Omega K(A, n{+}1) \simeq_* K(A, n)$, $K(h, n)$ for the map induced by a homomorphism h, and $F; G$ for composition of pointed maps (first F).

For pointed X, Y, $[X, Y]_*$ is the *set* of pointed homotopy classes, the set truncation of $\mathrm{Map}_*(X, Y)$; classes compose and form a category. Two classes are equal when their representatives are merely equal; no representative is ever chosen. Homotopy groups are $\pi_3 X = \pi_1(\Omega^2 X)$ and, by definition,

$$\pi_4 X = \pi_3(\Omega X) = \pi_1(\Omega^3 X),$$

functorial on maps and on classes (looping descends to classes since $[\Omega X, \Omega Y]_*$ is a set).

Axiom 1 (S^3)**.** A type S^3 with a point base and $\mathsf{surf} : \Omega^3(S^3, \mathsf{base})$ such that for every X, $f \mapsto (f\,\mathsf{base}, \mathsf{ap}^3_f\,\mathsf{surf})$ is an equivalence $(S^3 \to X) \simeq (x : X) \times \Omega^3(X, x)$.

The axiom is a hypothesis of the module `Sphere3` (its parameters `S3`, `base`, `surf`, `ump3`), not of the whole development, so only the results that take these parameters depend on it. From it one proves that the pointed evaluation $\mathrm{Map}_*(S^3, X) \to \Omega^3 X$, $F \mapsto (\Omega^3 F)(\mathsf{surf})$, is an equivalence, and that it is natural: for $G : X \to_* Y$, the evaluation of $F; G$ is $\Omega^3 G$ applied to the evaluation of F.

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2. The strategy: a marked universal property

The proof is a Yoneda argument on classes; the same argument, one level down, gives $\pi_3(S^2) = \mathbb{Z}$ (formalised first, `doc/pi3-s2-postnikov.pdf`).

Let $T = \|S^3\|_4$ with unit $\eta : S^3 \to_* T$. From the axiom, S^3 is 2-connected (into a 2-type Y evaluation at the base point is an equivalence, so $\|S^3\|_2$ is contractible), hence T is a 2-connected 4-type, $\eta^* : \mathrm{Map}_*(T, Y) \to \mathrm{Map}_*(S^3, Y)$ is an equivalence of pointed mapping types for every 4-type Y, hence a bijection $[T, Y]_* \to [S^3, Y]_*$ on homotopy classes, and $\pi_4\eta$ is an isomorphism (the fibres of $\Omega^k\eta$ are $(4-k)$-connected).

Proposition 2 (two instances suffice)**.** *Let P be a 4-type and $s \in [S^3, P]_*$ such that $s^* : [P, Y]_* \to [S^3, Y]_*$ is a bijection for $Y = P$ and for $Y = T$. Then $\pi_4 S^3 \cong \pi_4 T \cong \pi_4 P$.*

Proof. Let $u \in [T, P]_*$ with $[\eta]; u = s$ (η^* at P) and $t \in [P, T]_*$ with $s; t = [\eta]$ (s^* at T). Then $s; (t; u) = [\eta]; u = s = s; \mathrm{id}$, so $t; u = \mathrm{id}_P$; and $[\eta]; (u; t) = s; t = [\eta]$, so $u; t = \mathrm{id}_T$. Hence $\pi_4 u$ and $\pi_4 t$ are inverse isomorphisms. □

So the whole proof consists in constructing a 4-type P with $\pi_4 P = \mathbb{Z}/2$ and a class s with this universal property. The candidate is the delooping BR of $R = \mathrm{Fib}\bar{\kappa}$, the fibre of the cup-square k-invariant $\kappa : K(\mathbb{Z}, 2) \to_* K(\mathbb{Z}, 4)$ with its coefficient reduced modulo 2, $\bar{\kappa} = \kappa; K(\rho, 4)$ (§3). No result about S^2 is used.

3. The candidate BR

The cup square. Following Wärn's construction, the circle property of $K(\mathbb{Z}, 1)$ and delooping give a bipointed map $c_2 : K(\mathbb{Z}, 2) \to_* \mathrm{Map}_*(K(\mathbb{Z}, 2), K(\mathbb{Z}, 4))$, normalised so that looping it and evaluating at the generator of $\pi_1 K(\mathbb{Z}, 1)$ gives back the identity of $K(\mathbb{Z}, 2)$; the cup square is its diagonal, $\kappa(x) = c_2(x)(x)$ (`doc/cup-square.pdf`).

Cohomology of $K(\mathbb{Z}, 2)$. For every abelian group B:

- (H3) $[K(\mathbb{Z}, 2), K(B, 3)]_*$ is a point (from the circle property; `doc/circle-torsors.pdf`);
- (H4) $\Phi_B : \mathrm{Hom}(\mathbb{Z}, B) \to [K(\mathbb{Z}, 2), K(B, 4)]_*$, $h \mapsto [\kappa; K(h, 4)]$, is a bijection (`doc/h4.pdf`): every class is the diagonal of a unique bipointed map, and bipointed maps are classified by $g \mapsto c_2; K(g, 4)_*$.

With $P_2 = \mathrm{Fib}\kappa$ this gives the mapping theorem and $\pi_3(S^2) = \mathbb{Z}$.

Reduction mod 2. Let $\rho : \mathbb{Z} \to \mathbb{Z}/2$, $\bar{\kappa} = \kappa; K(\rho, 4) : K(\mathbb{Z}, 2) \to_* K(\mathbb{Z}/2, 4)$ and $R = \mathrm{Fib}\bar{\kappa}$, a simply connected 3-type with $\pi_2 R = \mathbb{Z}$ and $\pi_3 R \cong \mathbb{Z}/2$ as groups.

Theorem 3 (R deloops; `doc/cross-term.pdf`)**.** *There is $\hat{F} : K(\mathbb{Z}, 3) \to_* K(\mathbb{Z}/2, 5)$ with $\Omega\hat{F} = e_2; \bar{\kappa}; e_4^{-1}$. Hence $BR = \mathrm{Fib}\hat{F}$ is a pointed 2-connected 4-type with $\Omega BR \simeq_* R$.*

Idea. By Wärn's delooping criterion ([1], Lemma 9), $f = e_2; \bar{\kappa}; e_4^{-1}$ is a looping as soon as, for every loop p, f agrees with its translate $q \mapsto (fp)^{-1} \cdot f(p \cdot q)$; at these levels the space of such data is a proposition, so no coherence for the pointing in p is needed. The defect is the *cross term* "$\bar{\kappa}(x + y) - \bar{\kappa}(x) - \bar{\kappa}(y)$", a bipointed map; by the classification of (H4) it vanishes as soon as its diagonal "$\bar{\kappa}(2x) - 2\bar{\kappa}(x)$" does, and both terms of the diagonal vanish by construction because $2 = 0$ in $\mathbb{Z}/2$: doubling on $\Omega K(\mathbb{Z}/2, m{+}1)$ is $K(2, m) = K(0, m) = 0$, and precomposing c_2 with $K(2, 2)$ doubles its invariant. □

4. The mapping theorem for BR

A *level-3 triple* is (A, B, K) with $K : K(A, 3) \to_* K(B, 5)$; its fibre $\mathrm{Fib}K$ is a pointed 2-connected 4-type, with projection $q : \mathrm{Fib}K \to_* K(A, 3)$.

Theorem 4 (lifting)**.** *For a pointed Q, suppose (E) every $F : Q \to_* K(A,3)$ has merely $F;K = 0$, and (U) every $d : Q \to_* \Omega K(B,5)$ is merely 0. Then $q_* : [Q, \mathrm{Fib}K]_* \to [Q, K(A,3)]_*$ is a bijection.*

This is the lifting lemma of the mapping theorem for Fibκ (`doc/mapping-fib.pdf`) one level up. It is applied to the two sources $Q = S^3$ and $Q = BR$. For S^3 both conditions are immediate (pointed maps out of S^3 are 3-loops, and $\Omega^3 K(B,5)$, $\Omega^4 K(B,5)$ are connected).

(U) for BR. Looping is an equivalence $\mathrm{Map}_*(BR, \Omega K(B,5)) \simeq \mathrm{Map}_*(R, \Omega^2 K(B,5))$, and $[R, K(B,3)]_* = 0$: a map out of $R = \mathrm{Fib}\bar{\kappa}$ restricts on the fibre to a map that deloops to some $K(g,4)$, the fibre sequence forces $[\bar{\kappa}; K(g,4)] = 0$, and

$$\Psi_B : \mathrm{Hom}(\mathbb{Z}/2, B) \to [K(\mathbb{Z},2), K(B,4)]_*, \qquad g \mapsto [\bar{\kappa}; K(g,4)] = \Phi_B(\rho; g)$$

is injective by (H4); then (H3) kills what remains (`doc/fibre-h3.pdf`, with a fibre coefficient group).

(E) for BR: the decisive point. A map $F : BR \to_* K(A,3)$ factors merely through $p : BR \to_* K(\mathbb{Z},3)$ as $F = p; G$ with $G : K(\mathbb{Z},3) \to_* K(A,3)$ (the fibres of p are 3-connected). Let $M = G; K : K(\mathbb{Z},3) \to_* K(B,5)$; then merely $F;K = p;M$, and the obstruction is the class of $H = \mathrm{unconj}(\Omega M) : K(\mathbb{Z},2) \to_* K(B,4)$, i.e. $h \in \mathrm{Hom}(\mathbb{Z}, B)$ with $\Phi_B(h) = [H]$. The fibre sequence of BR kills it only if h factors through ρ, i.e. if $2h = 0$. So one must show that *for every target the obstruction is 2-torsion.* The point is that H is the looping of a map *by construction*:

Lemma 5 (necessity)**.** *If $\Phi_B(h) = [\mathrm{unconj}(\Omega \hat{G})]$ for some $\hat{G} : K(\mathbb{Z},3) \to_* K(B,5)$, then $2h = 0$.*

Idea. A looping commutes with squaring of loops, and squaring is $K(2,2)$ on the source and $K(2_B,4)$ on the target; so $[K(2,2)];[H] = [H];[K(2_B,4)]$. A computation with the bipointed classification gives $(2x)^2 = 4x^2$, i.e. $[K(2,2)];[\kappa] = [\kappa];[K(4,4)]$; its only new ingredient is the naturality of the invariant of a bipointed map under precomposition in the second variable. Hence $\Phi_B(4h) = \Phi_B(2h)$, and $2h = 0$ since Φ_B is injective. □

This is the central reason why $\mathbb{Z}/2$ appears: looping forces doubling to act *linearly* on the obstruction ($h \mapsto 2h$), whereas the cup square makes it act *quadratically* ($h \mapsto 4h$); comparing the two forces $2h = 0$.

Then $h = \rho; g$, so merely $H = \bar{\kappa}; K(g,4)$, $\Omega M = \Omega(\hat{F}; K(g,5))$, and since looping is injective on maps $K(\mathbb{Z},3) \to_* K(B,5)$, merely $M = \hat{F}; K(g,5)$ and $F;K = p;M = (p;\hat{F});K(g,5) = 0$. This is why the target is presented one level up, with a k-invariant $K(A,3) \to K(B,5)$: every obstruction is then a looping, and no coherence cell has to be normalised.

The marked class. From the axiom and the circle property one builds $\sigma_3 : S^3 \to_* K(\mathbb{Z},3)$ with $(\Omega^3\sigma_3)(\mathsf{surf})$ the generator; σ_3^* is an equivalence on maps into 3-types (a composite of five equivalences). Let

$$s = p_*^{-1}[\sigma_3] \in [S^3, BR]_*$$

(Theorem 4 at $Q = S^3$ for the triple $(\mathbb{Z}, \mathbb{Z}/2, \hat{F})$). Then s^* is a bijection into every $K(A,3)$, and the commuting square of q_* for the two sources gives:

Theorem 6 (mapping theorem)**.** *For every level-3 triple, $s^* : [BR, \mathrm{Fib}K]_* \to [S^3, \mathrm{Fib}K]_*$ is a bijection; hence so is $s^* : [BR, Y]_* \to [S^3, Y]_*$ for every Y merely pointed equivalent to some* Fib$K$*.*

5. Presentations of 2-connected 4-types

Theorem 7. *Every pointed 2-connected 4-type Y is merely pointed equivalent to* Fib$K$ *for a level-3 triple (A, B, K).*

Idea. As the presentation of $\|S^2\|_3$ (`doc/postnikov-presentation.pdf`), one level up. Let $V = \|Y\|_3$; it is $K(A,3)$ with $A = \Omega^3 V$ (a 2-connected 3-type is recognised by delooping its loop space twice). The fibres E_z of $Y \to V$ are 3-connected 4-types, hence $K(B,4)$ with $B = \Omega^3(\Omega E_{v_0})$. The structures $K(B,4) \simeq_* E_z$ form a set over Y, and a section through the given one exists since Y is 2-connected; so every E_z carries a homogeneous structure, i.e. is a $K(B,4)$-torsor, and $z \mapsto E_z$ is a map $k' : V \to K(B,5)$ with $\mathrm{Fib}k' \simeq Y$. Take $K = e_V; k'$; the pointing is only mere since Y is connected. □

The group B is not identified: for $Y = T$ it is $\pi_4 T$, the unknown. Since the goal "s^* is a bijection" is a proposition, the mere presentation of T suffices.

6. Conclusion

Theorem 8. *Under Axiom 1, $\pi_4(S^3) \cong \mathbb{Z}/2$ as groups.*

Proof. Theorem 6 applies to $Y = BR = \mathrm{Fib}\hat{F}$ directly and to $Y = T$ through Theorem 7. Proposition 2 with $P = BR$ gives

$$\pi_4 S^3 \cong \pi_4 T \cong \pi_4 BR = \pi_3(\Omega BR) \cong \pi_3 R \cong \mathbb{Z}/2.$$ □

7. The formalisation

The proof is checked in Unit, in `sag/pi3-s2-postnikov` (`./segc sag/pi3-s2-postnikov all`: 83 modules; a fresh check, 4038 declarations, 0 failed, about 2 s). The final declaration is

```
Pi4Sphere.Pi4S3[S3, base, surf, ump3].pi4-S3 :  GIso[pi4 S3 base, ab-grp Z2ab]
```

whose only parameters beyond the foundations (univalence, images and their surjectivity) are the S^3 axiom; the 4-truncation is constructed inside. The S^2 axiom is not used.

`CupSquare`, `H3Proof`, `H4Proof`	the cup square, (H3), (H4)
`CrossTerm`, `DeloopR`	the cross term vanishes; $\hat{F}$, BR, $\Omega BR \simeq R$
`Necessity`, `FibreH3`	Lemma 5, $(2x)^2 = 4x^2$; $[R, K(B,3)]_* = 0$
`MapBR`	Theorem 4; (E) and (U) for BR
`Sphere3Axiom`, `SourceS3`	the axiom; σ_3; s; Theorem 6
`Sphere3Trunc`	S^3 is 2-connected; $T = \|S^3\|_4$; π_4 on classes
`Presentation4`	Theorem 7; the mapping theorem at T
`Pi4Sphere`	Proposition 2; Theorem 8

For every declaration the checker reports which definitions it had to unfold; the `.opens` files of the development record, declaration by declaration, the definitions allowed to be unfolded, and the check against them passes for every declaration. These specifications were generated from the checker's report and reviewed by kind (which kinds of unfolding each text step justifies); the textual audit, entry by entry against the notes, is not yet complete. The later parts (the cup square, (H3), (H4), the fibre lemma, the mapping theorem for Fibκ, the presentation of $\|S^2\|_3$, and everything from the cross term onwards) were written from an informal note first; the earliest modules (the sphere axiom, homotopy classes and groups, the realisation of Postnikov triples, finite truncations) were written before their notes. Detailed notes: `br-universal.pdf` (this proof in full), `cross-term.pdf`, `cup-square.pdf`, `h4.pdf`, `fibre-h3.pdf`, `circle-torsors.pdf`, `mapping-fib.pdf`, `postnikov-presentation.pdf`, `finite-truncation.pdf`, and `pi3-s2-postnikov.pdf` for $\pi_3(S^2) = \mathbb{Z}$.

References

[1] D. Wärn. *Eilenberg–MacLane spaces and stabilisation in homotopy type theory.* arXiv:2301.03685, 2023.